

Presentation Script

HCI Group Project Presentation Script

Project: Heuristic Evaluation and Redesign of the 12306 Home Ticket-Search Flow

Presenters:

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Suggested total time: 5 to 7 minutes.

Speaker 1: You Shengxuan

Good morning. Our group project is a heuristic evaluation and redesign of the China Railway 12306 home ticket-search flow.

We selected 12306 because it is a high-frequency public service system. Many users open the app with one urgent goal: search a train ticket and continue to booking. The real screenshot on our poster shows that the home page already contains the correct train-ticket search card, but it is surrounded by advertising, unrelated search suggestions, many service shortcuts, announcements, and bottom navigation.

Our representative task is to search a one-way train ticket from Hefei South to Shengzhou Xinchang on June 10. Our evaluation objective is to check whether the first screen supports this task clearly, and then propose a UI-only redesign based on Nielsen's usability heuristics.

Speaker 2: Xu Leyu

I will explain the evaluation method and task flow.

We used Nielsen's ten usability heuristics as our framework. The most relevant heuristics in this case are visibility of system status, consistency and standards, recognition rather than recall, flexibility and efficiency of use, aesthetic and minimalist design, error prevention, and help and documentation.

The task flow has five steps. First, the user lands on the 12306 home page and sees the large promotion and search area. Second, the user must stay in the train-ticket tab instead of choosing flight or bus ticket. Third, the user checks the departure station, arrival station, date, and student-ticket option. Fourth, the user taps the query button and may notice history routes. Fifth, the user continues to the train-list page and later completes booking.

For each step, we checked whether the interface gives clear feedback, whether the main action is visually dominant, and whether the user has to remember or infer information. We then assigned

severity levels: cosmetic, minor, major, or catastrophic.

Speaker 3: Wang Yihang

I will summarize the main usability problems in the current interface.

The first major issue is advertising dominance. The top reward banner uses a strong red background and large text, so it receives more attention than the ticket-search task. This violates aesthetic and minimalist design.

The second issue is fragmented intent. On the same screen, the user sees financial search, pet shipping, floating utility icons, train, flight, bus, service shortcuts, announcements, and bottom navigation. These functions are useful, but showing all of them at the same time weakens the primary path.

The third issue is the service shortcut grid. Some services are travel-related, such as station display and timetable, but hotels, shopping, insurance, tourism, and rental cars appear before the user finishes searching tickets. This increases cognitive load before the main task is completed.

The fourth issue is weak contextual help. The student-ticket checkbox is visible, but eligibility is not explained at the decision point. Announcements are visible, but users cannot immediately know whether they affect the current ticket search.

Speaker 4: Wang Yibin

I will explain our redesigned home page prototype.

The redesign keeps the real 12306 home-page structure, but changes visual priority. First, the advertisement is reduced and moved behind a secondary promotion entry. This makes the train-ticket card the strongest visual object on the screen.

Second, the main form is protected. Departure, arrival, date, student option, and query button are grouped together as one clear action cluster. This supports recognition rather than recall and reduces the chance of choosing the wrong route or date.

Third, we use progressive disclosure. Core travel tools, such as timetable, station display, and warm service, remain visible. Commercial or less urgent services, such as hotels, shopping, insurance, rental cars, and tourism, move under a More Services entry until the user needs them.

Fourth, contextual decision support is added. Student-ticket rules appear next to the checkbox, and announcements only interrupt the flow when they affect booking, payment, or ticket validity.

Speaker 5: Liao Jisong

I will conclude with the expected impact and evaluation metrics.

Our redesign does not remove 12306 functions. It reorganizes attention so that the user can complete the ticket-search task first, while secondary services remain available after the main goal is

protected.

The expected UX impact is lower cognitive load, clearer task priority, fewer wrong-route or wrong-date mistakes, and better support for first-time users and older adults. The interface should feel less crowded because the most important action is visually separated from advertisements and secondary services.

The redesign can be evaluated with measurable criteria: time to first query, number of wrong-route or wrong-date errors, task completion rate, and perceived workload. If these metrics improve, the redesign has successfully improved usability.

In conclusion, our proposal is not just a visual cleanup. It is a task-first redesign based on HCI principles and Nielsen's heuristics. Thank you.